Extremal mass of equal-spin EGB black holes in $D = 5$: weak series to s^6, its singularity, and the strong-coupling limit

16–23 October 2026. Continuation of `extremal-mass-report.pdf`. $G_5 = 1$, action $R + \alpha\mathcal{L}_{\rm GB}$, $J_1 = J_2 = J$, $M_{\rm ext} = \pi^{1/3} J^{2/3} m(s)$, $s = \pi^{2/3}\alpha/J^{2/3}$.

Summary

1. **Weak coupling, two more exact orders.** d_5 and d_6 (below) from the exact exterior program; the horizon shifts X_5, X_6 and the first law are checked at each order.

2. **The near-horizon data are algebraic.** The extremal throat equations fix $g_H(s)$ exactly; their series reproduces the perturbative horizon through α^4 and predicted X_5, X_6 before the exterior computation finished.

3. **Radius of the weak series.** $g_H(s)$ has its nearest branch points at $s_* = 0.0059232 \pm 0.0548189\,i$, $|s_*| = 0.05514$, a square-root fold of the throat. The signs of $d_1, \ldots, d_6$ are those of $\mathrm{Re}\, s_*^{-n}$ (6 of 6). Conjecture: $m(s)$ has the same singularity, so the weak series converges only for $|s| < 0.05514$.

4. **Strong coupling from pure-Lovelock universality.** Static $D = 5$ EGB is the "derivative-degenerate" Lovelock theory of Dadhich–Pons–Prabhu (arXiv:1201.4994); its small-hole core is a BTZ black hole with $\ell^2 = 4\alpha$, $1/(4G_3) = 3\pi\alpha$. Using the extremal-BTZ relation $\Delta M = S^2/(48\pi^3\alpha^2)$ with the *exact* extremal entropy gives
$$M_{\rm ext} = \frac{3\pi\alpha}{4} + \frac{4J}{3\sqrt{\alpha}} + \ldots, \qquad m(s) = \frac{3s}{4} + \frac{4}{3}s^{-1/2} + \ldots$$
which reproduces the coefficient $4/3$ of the earlier matched-asymptotics candidate by an independent route.

5. **Numerics at strong coupling.** Two-sided shooting ($s \le 8.6$) and multiple shooting with quadruple-precision near-horizon segments reach $s = 103$ (the spectral branch stopped at 5.24). Fitting $(m - 3s/4)\sqrt{s} = a_1 + a_2 s^{-1/2} + a_3 s^{-1} + \ldots$ (5–7 terms, $s \ge 10, 15, 20$) gives $a_1 \in [1.33330, 1.33369]$ with all coefficients free; with $a_1 = 4/3$ fixed, $a_2 \in [0.45658, 0.45669]$ (conditional prediction 0.456527) and $a_3 \in [-0.268, -0.265]$. Fixing also $a_2 = 0.456527$ (six further terms) fits as well as the free fits (rms $\le 1 \times 10^{-8}$) and gives $a_3 \in [-0.2625, -0.2623]$. Hence, numerically,
$$m(s) = \frac{3s}{4} + \frac{4}{3}\, s^{-1/2} + a_2\, s^{-1} + a_3\, s^{-3/2} + O(s^{-2}), \qquad a_2 = 0.4566 \pm 0.0002, \quad a_3 \simeq -0.26,$$
confirming $a_1 = 4/3$ and the conditional matched-asymptotics value $a_2 = 0.456527$ (fits with a_2 fixed to it are as good as free fits).

6. **No closed form.** $\zeta(3), \zeta(5), \zeta(3)^2, \pi^{2k}$ in d_n exclude an algebraic $m(s)$; the complex singularity at $|s| = 0.055$ excludes a simple elementary form built from the two ends. A ten-term approximant with the exact ends (§4) reproduces all 126 numerical points to 1.1e-05 (relative).

1. Weak coupling through s^6

$$m(s) = \tfrac{3}{2} + s - \tfrac{236}{105}s^2 + d_3 s^3 + d_4 s^4 + d_5 s^5 + d_6 s^6 + O(s^7),$$

$$d_5 = \tfrac{447853140007115845664}{6100990520625} + \tfrac{657983488}{175}\zeta(3) - \tfrac{3972366336}{625}\zeta(2)^2 - \tfrac{1464139776}{25}\zeta(5) = 423.920197004910\ldots$$
$$d_6 = -\tfrac{12452546090281502179186501658 24}{88766575114501659375} + \tfrac{15735548018688}{6125}\zeta(2)^3 - \tfrac{57616520085504}{109375}\zeta(2)^2 + \tfrac{5569511424}{35}\zeta(3)^2$$
$$- \tfrac{94884252024832}{165375}\zeta(3) + \tfrac{2699634147328}{625}\zeta(5) = -4737.494826989936\ldots$$

(d_3, d_4 as in the earlier report.) Method and conditions are those of `code/pert.py`: exact hyperlogarithm quadratures, constants fixed by asymptotic flatness and a double zero of g^{rr}; exact *conditional* on the PSLQ identification (100 digits) of the regularised horizon constants C_w with multiple zeta values. At orders 5 and 6 the first-law residual $dM = 2\Omega_H\, dJ$ is identically zero, and the horizon shifts $X_5 = 292028416/729$, $X_6 = 58879115264/6561$ equal the throat predictions of §2. Run times 25.5 min (order 5) and 1 h 38 min (order 6).

2. Algebraic near-horizon data and the radius of convergence

With $a = \alpha/g_H$ and $q = h_H/g_H$ the extremal throat satisfies $P(a,q) = 16a^2q^2(4-q) - 8a(2q^2-7q+4) + q - 2 = 0$ and $8J^2/(\pi^2 g_H^3) = q^2(4-q)[1 + 4a(4-3q)]/8$. At fixed J these give $g_H(s)$ as an algebraic function, with $g_H = 1 + \frac{8}{3}s + \frac{16}{9}s^2 - \frac{10496}{81}s^3 - 512s^4 + \frac{9125888}{729}s^5 + \dots$ (in units where $g_H(0) = 1$). Two independent checks: (i) these coefficients equal the horizon shifts of the exterior computation through s^6; (ii) the map $s(q)$ reproduces the coupling of every point of the earlier spectral branch from its q alone, to 10^{-6}.

Eliminating (g_H, q) from the throat equations and their Jacobian gives the candidate branch points; Padé approximants of 60 exact Taylor coefficients select those on the physical sheet: $s_* = 0.0059232 \pm 0.0548189\,i$. In the q-plane s_* is a critical point of the explicit map $s(q)$, at $q_c = 2.16110 \mp 0.94332\,i$: a square-root fold. The Padé poles beyond s_* lie on the line $\mathrm{Re}\, s \simeq 0.006$, i.e. a cut running away from s_*.

Evidence that $m(s)$ shares it. $\operatorname{sign} d_n = \operatorname{sign} \mathrm{Re}\, s_*^{-n}$ for $n = 1, \dots, 6$ ($\arg s_* = 83.8°$, so the pattern is not trivial); $|d_n||s_*|^n$ decreases slowly. A conformal map that opens the cuts from $s_*, \bar{s}_*$ turns the divergent series into one oscillating around the numerical $m(s)$ out to $s \approx 0.5$ (`series/conformal.py`); its slow convergence indicates further singularities near the boundary of the mapped disk. None of this is a proof.

3. Strong coupling: pure-Lovelock universality

For $D = 5$ EGB the static master equation is $\psi + 2\alpha\psi^2 = \mu/r^4$, $V = 1 - \psi r^2$; completing the square it is the $N = 2$ "derivative-degenerate" polynomial $F(\psi) = -a + (\psi - \beta)^2$ of arXiv:1201.4994 (their eq. 15) with $\beta = -1/(4\alpha)$. For $r \ll \sqrt{\alpha}$ it behaves like pure Gauss–Bonnet in its critical odd dimension, which (their §IV) has BTZ-like, universal thermodynamics: with $\mu = r_h^2 + 2\alpha$,

$$\Delta M = \tfrac{3\pi}{8} r_h^2, \quad S = 6\pi^2 \alpha\, r_h, \quad T = \frac{r_h}{8\pi\alpha} \iff \text{BTZ with } \ell^2 = 4\alpha,\ \tfrac{1}{4G_3} = 3\pi\alpha.$$

The critical static solution $\mu = 2\alpha$, $M = 3\pi\alpha/4$, is the massless BTZ point. For an extremal BTZ hole with the same (ℓ, G_3), $\Delta M = 4G_3 S^2/(4\pi^2\ell^2) = S^2/(48\pi^3\alpha^2)$. The extremal entropy is known exactly from the throat; in $t = (2a)^{-1/2}$

$$\frac{S_{\rm ext}^2}{48\pi^3\alpha^{3/2}J} = \frac{4}{3} + \frac{3}{8}t^2 + O(t^3) \quad \Longrightarrow \quad \Delta M_{\rm ext} = \frac{4J}{3\sqrt{\alpha}}\,(1 + O(t^2))$$

(`strong/btz_universality.py`, symbolic). The step that is assumed, not derived, is that the rotating extremal core obeys the static BTZ dictionary. The absence of an $O(t)$ term means the next coefficient a_2 of s^{-1} in $m(s)$ cannot come from the core.

4. Numerical solution at large s

Methods. (i) $s \le 5.24$: the spectral branch of `work/kkr-extremal`. (ii) $2 \le s \le 8.6$: two-sided shooting (`strong/twoside.py`): outward from the horizon, with its formal series (rank-one level blocks) plus the non-analytic throat mode $B\,x^\gamma$ (γ the exact throat exponent), inward from the large-r series with free mass μ and squashing amplitude b_4; $m = 3\mu/4$. (iii) $s \ge 9$: multiple shooting (`strong/ms.py`, `ms_cont.py`). Single shooting fails for $s \gtrsim 10$ because in the core the squashing mode behaves as $e^{\pm 4\sqrt{6\alpha}/r}$ (a factor e^{76} at $\alpha = 10$). The solver places nodes at equal WKB phase in the core, integrates segments with $x \le 0.5\rho$ in quadruple precision (`__float128` Bulirsch–Stoer) from 60-digit horizon series, and continues in α across the resonances $\gamma = 3, 4, 5$ ($\alpha = 4.83, 13.97, 31.66$). The large-$r$ data (series in $1/r$ with free μ, b_4) are imposed at $R = 12\sqrt{\alpha}$ ($s < 80$) and $R = 8\sqrt{\alpha}$ ($s > 80$). At larger R the squashing datum b_4 enters $h(R)$ only at relative order R^{-6}; Newton then stalls on a residual floor and m is wrong by up to $\sim 10^{-6}$ (a first run with $R = 25\sqrt{\alpha}$ is discarded for this reason; `ms_check.py`).

Checks. At the common point $s = 2.98202$ the two-sided and spectral values differ by 3.4e-07; against a spline of the spectral branch the shooting/BVP values deviate by at most 2.0e-05, the size of the spectral branch's own constraint violations (median 3e-06, max 7e-05). Varying the shooting parameters moves m by at most 2.8e-08. Multiple-shooting Newton residuals are below 3e-10. A degree-6 polynomial through the 18 multiple-shooting points with $9.16 \le s \le 10.3$, extrapolated to the BVP point $s = 9.0037$, agrees with it to 5e-09. At $s = 48.9$, $R = 8\sqrt{\alpha}$ and $12\sqrt{\alpha}$ agree to 5×10^{-10} and m is independent of the number of far nodes to 5×10^{-15}. Quality check on every point: αb_4 varies smoothly (for $s > 40$ it lies in $[-0.2255, -0.2214]$).

s	$m(s)$	$(m-3s/4)\sqrt{s}$	method
2	2.4446986200	1.33600560	shoot
2.4	2.6802239675	1.36363711	shoot
2.98202	3.0410066937	1.38923766	shoot
3.9	3.6394756194	1.41097629	shoot
5.07	4.4346318566	1.42334984	shoot
6.591	5.4998298751	1.42890434	shoot
8.5683	6.9145757941	1.42948377	shoot
9.00366	7.2290351282	1.42916958	BVP
9.16046	7.3425005931	1.42903362	MS
9.32	7.4580467159	1.42888461	MS
9.52	7.6030391456	1.42868400	MS
9.59986	7.6609741162	1.42859997	MS
9.86	7.8498672737	1.42831221	MS
10.08	8.0097946907	1.42805387	MS
11.0299	8.7020175250	1.42681968	MS
13.5121	10.5212075510	1.42316671	MS
16.5529	12.7633325407	1.41866050	MS
20.278	15.5223943713	1.41364796	MS
24.8414	18.9136032782	1.40838926	MS
32.562	24.6670488835	1.40131488	MS
39.8898	30.1384092467	1.39612703	MS
48.8667	36.8490546261	1.39113327	MS
59.8638	45.0770724775	1.38638522	MS
78.4693	59.0078117196	1.38048665	MS
96.1283	72.2365798289	1.37640334	MS
102.857	77.2785158831	1.37510768	MS

Strong-coupling fits ($\sigma = s^{-1/2}$; free: leading coefficients of $1, \sigma, \sigma^2, \ldots$; $a_1 = 4/3$: of $\sigma, \sigma^2, \ldots$ after subtracting $4/3$; rms of the fit to $(m - 3s/4)\sqrt{s}$):

fit	$s_{\min}$	terms	leading coefficients	rms
free	10	5	1.33369, 0.44648, -0.15638, -1.20273	9e-07
free	10	6	1.33334, 0.45635, -0.26292, -0.65016	3e-08
free	10	7	1.33331, 0.45763, -0.28038, -0.52680	1e-08
$a_1 = 4/3$	10	5	0.45661, -0.26572, -0.63587, -0.72216	4e-08
$a_1 = 4/3$	10	6	0.45669, -0.26776, -0.61457, -0.82906	2e-08
free	15	5	1.33351, 0.45100, -0.19687, -1.04730	2e-07
free	15	6	1.33333, 0.45688, -0.26917, -0.61436	9e-09
free	15	7	1.33333, 0.45683, -0.26841, -0.62052	9e-09
$a_1 = 4/3$	15	5	0.45664, -0.26621, -0.63186, -0.73601	1e-08
$a_1 = 4/3$	15	6	0.45659, -0.26466, -0.65039, -0.62776	9e-09
free	20	5	1.33345, 0.45268, -0.21350, -0.97594	6e-08
free	20	6	1.33333, 0.45682, -0.26833, -0.61983	1e-08
free	20	7	1.33330, 0.45789, -0.28617, -0.46370	9e-09
$a_1 = 4/3$	20	5	0.45662, -0.26571, -0.63673, -0.71557	1e-08
$a_1 = 4/3$	20	6	0.45658, -0.26457, -0.65151, -0.62130	1e-08

Global approximant (`strong/global2.py`). With $t = (1 + s/\kappa)^{-1/2}$,

$$m(s) \approx \frac{3s}{4} + \sum_{k=1}^{10} c_k\, t^k,$$

which is analytic at $s = 0$ and expands in $s^{-1/2}$ at infinity. Imposed exactly: $m(0) = 3/2$, $m'(0) = 1$, $c_1\sqrt{\kappa} = 4/3$, $c_2\kappa = 0.456527$; the other six c_k and $\kappa = 0.3562$ are fitted. Maximum relative error 1.1e-05 on $0 < s \leq 103$. Coefficients: 2.233910, 1.281505, -0.713812, 6.055449, -75.189708, 230.148117, -352.252482, 303.906361, -142.214324, 28.244984. It is an interpolant, not a closed form; it does not reproduce the complex singularity at $|s_*| = 0.055$, so its Taylor coefficients beyond s^1 are not those of $m(s)$.

5. What is and is not established

Exact: d_5, d_6 (conditional on the zeta identifications); the algebraic throat and $g_H(s)$; the location of s_*; $a_1 = 4/3$ *given* the BTZ dictionary for the rotating core. **Numerical:** $m(s)$ for $0 < s \leq 103$ with the error budget of §4;

$a_1 = 4/3$ to 4×10^{-4} (free fits); $a_2 \in [0.4566, 0.4567]$ with a_1 fixed, consistent with the conditional analytic value 0.456527 (supported, not derived here); $a_3 \in [-0.268, -0.262]$. **Conjectural:** that s_* is the nearest singularity of $m(s)$. **Not obtained:** a closed form for $m(s)$.

Code: `extremal-mass/code` (exact series), `extremal-mass/series`, `extremal-mass/strong`; resume notes `extremal-mass/RESUME.txt`.